APARUP SANTRA

Programming | Continuous Learning
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SUMMARY

Passionate MCA student with a strong interest in software development. Skilled in React.js, JavaScript, HTML, CSS, and SQL, with hands-on project experience. Enjoy building user-friendly applications, solving problems, and learning new technologies. Eager to grow as a developer and contribute to real-world projects.

PROJECTS

SpendWise: Expense Tracker

- Developed a responsive React expense tracker that lets users add, edit, and delete daily transactions and monitor their spending through an interactive dashboard.
- Implemented search, category and date-range filtering, sorting, and daily, monthly, and total summaries, with category-wise and six-month spending charts.

Tools: React.js, JavaScript, Tailwind CSS

Jago Guru: Edge-Based Wake Word Detection

- Developed an on-device Bengali wake-word detection system using ESP32-S3 and an INMP441 microphone for real-time audio recognition.
- Created an audio dataset and machine learning pipeline using Python, generating Log-Mel Spectrogram features and training a compact quantized CNN model.
- Integrated TensorFlow Lite for Microcontrollers with VAD and multi-frame prediction logic to improve detection reliability and provide LED-based feedback.

Tools: Python, C/C++, Tenserflow, Numpy, ESP32-S3, ESP-IDF.

HealthMate: Wellness Tracking Application

- Developed a mobile wellness application with React Native and Expo to help users track daily activity, monitor health progress, and build healthy routines through gamification.
- Implemented exercise and hydration tracking, work session timers, health statistics, achievements, streaks, leaderboards, and peer challenges, supported by a Node.js, Express.js, and MongoDB backend.

Tools: React Native, Expo, TypeScript, NativeWind, Node.js, Express.js, MongoDB, Bun

EDUCATION

Master of Computer Applications, Brainware University — **Expected July 2027**

Bachelor of Computer Applications, Brainware University — **08/2022 - 2025**
Percentage: 82.22%

Higher Secondary (XII), Commerce, Paniparul Mukteswar High School — **08/2020 - 2022**
Percentage: 88%

KEY ACHIEVEMENTS

- **Hackathon Achievement:** TEXIBITION 2026: Secured **5th place** in a college-level hackathon by collaborating with a team to develop and present a technical solution.
- **Sports Achievement:** Represented at the district level in Football and Badminton (Under-14 and Under-16 categories) and at the state level in Football (Under-16 category).

ADDITIONAL INFORMATION

Frontend: React.js, React Native, JavaScript, HTML, CSS.
Programming and Database: Python, SQL, Firebase,MongoDB
Tools and Platforms: Git, GitHub, Linux, Power BI, Tableau, Expo, Bun
Languages: Bengali, Hindi, English
Certifications: Python — Udemy; Data Analytics and Visualization — Accenture ; Data Science - AICT PVT LTD